

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : D7

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1. Problem Title: Online Auction System

2. Problem Description:

An **Online Auction System** is a web-based platform where sellers can list items for sale and buyers can place bids on those items. Users can register, log in, view available auctions, and bid on products within a specified time. The system tracks bids and determines the highest bidder when the auction ends, declaring them the winner. It also maintains records of users, items, and bids.

3. Identified Entities (Classes) Class

Name	Description
Seller	The Seller class represents users who register in the system to list items for auction and manage the sale of their products.
Buyer	The Buyer class represents users who browse available items and place bids on products they want to purchase.
Admin	The Admin class represents the system administrator responsible for managing users, approving products, and monitoring auction activities.
Item	The Item class represents the products listed by sellers for auction along with their details such as name, description, and starting price.
Bid	The Bid class represents the bids placed by buyers on items during the auction process.
Auction	The Auction class manages the bidding process for an item and determines the winner when the auction ends.
User	The User class contains registration and login part and from there sellers and buyers are classified.
Database	The Database class stores input information and id counters.

Class Name	Attributes	Methods(behaviours)
Seller	1.Name(String) 2. Password(String) 3. personalInfo(Arraylist)	1. Register() 2.Login() 3. addItem() 4.auctionClose() 5. contactBidder()
Buyer	1.Name(String) 2. Password(String) 3. personalInfo(Arraylist)	1. Register() 2. Login() 3. viewItem() 4. placeBids() 5. viewResults()
Admin	1. adminId 2. Password	1. manageMembers() 2. approveProducts() 3. viewBidDetails()
Item	1.itemId 2.itemName 3.description 4.startingPrice 5.currentBid 6.auctionEndTime 7.sellerId	1.displayDetails() 2. updateBid()
Bid	1.bidId 2.bidAmount 3.bidTime 4.ItemId	1. placeBid() 2. getHighestBid()

Auction	1. auctionId 2. item 3. startTime 4. endTime 5. highestBid	1.startAuction() 2. closeAuction() 3. declareWinner()
User	1. name 2. password 3. personalInfo	1. register() 2. login()

